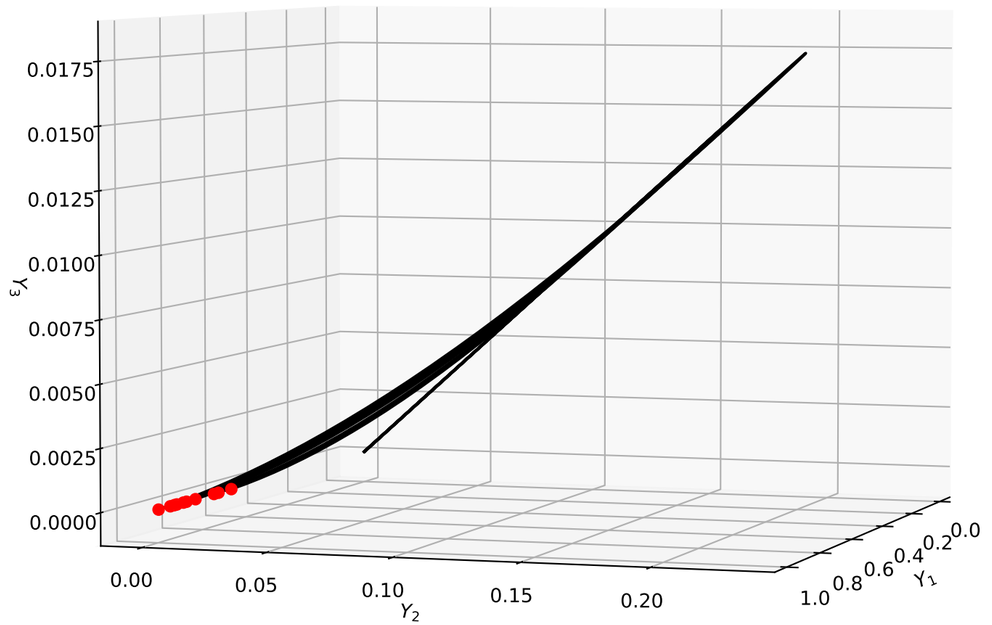Variable Composition: $X_{O_2} \in [0.5, 1.0]$